

Run 06 | Business exit

India

IV + controls + NES

Target Entrepreneurial/business exit in the past 12 months; binary 1 vs. 0
Sample 27,106 respondents **Coverage** 1 country; 2015–2022
Positive cases 1,402 (5.17%) **Case time** 1m 03s
Train / val. / test 18,434 / 3,252 / 5,420 (68% / 12% / 20%, stratified)
SCT predictors (12): Self-efficacy, Opportunity perception, Fear of failure, Knowing entrepreneurs, Entrepreneurial intention, Media coverage, Business angel, Established business owner, Good career choice, Ease of starting a business, High status, Social enterprise.
Legatum predictors (13): Prosperity index score, Safety and security, Personal freedom, Governance, Social capital, Investment environment, Enterprise conditions, Market access and infrastructure, Economic quality, Living conditions, Health, Education, Natural environment. Total independent variables: 25.
Controls (10): Survey year, Age, Gender, Education attainment, Household income, Household size, Occupational status, GDP per capita (PPP), Population, GDP growth. Country and respondent ID are excluded as predictors.
NES controls (12): Financing, Government support, Taxes and bureaucracy, Government programs, School entrepreneurship education, Higher/vocational entrepreneurship education, R&D transfer, Commercial/legal/professional services, Market changes, Market entry, Physical infrastructure, Cultural and social norms. Total predictors: 47.
Protocol: Ten CPU models; model seed 20260925; fixed splits by sample/outcome; training-only preprocessing; F1 threshold selected on validation data. Test scores below are unweighted.

Result in brief

Highest observed ROC-AUC: H2O GBM (0.8490). Highest F1: H2O GBM (0.3755). For H2O GBM, adding NES changed ROC-AUC by +0.0072 on the same test records.

Model performance on held-out test records

Model	ROC-AUC	PR-AUC	Acc.%	Prec.%	Recall%	F1	Time
H2O GBM	0.8490	0.3945	94.35	43.81	32.86	0.3755	5s
CatBoost	0.8351	0.3730	94.94	52.03	27.50	0.3598	25s
XGBoost	0.8344	0.3596	93.71	36.20	28.57	0.3194	<1s
Random Forest	0.8341	0.3788	93.82	38.30	32.14	0.3495	2s
Lasso	0.8121	0.2980	92.12	27.38	31.79	0.2942	5s
Elastic Net	0.8120	0.2983	92.05	27.05	31.79	0.2923	5s
Logistic Regression	0.8109	0.3175	93.10	33.09	32.86	0.3297	<1s
Ridge	0.8098	0.2940	94.17	39.41	23.93	0.2978	10s
Naive Bayes	0.7531	0.1683	93.21	26.60	17.86	0.2137	8s
Decision Tree	0.6046	0.1937	94.67	45.95	18.21	0.2609	<1s

Models ranked by ROC-AUC; small differences are descriptive, not significance tests. Time includes fit, prediction and output processing. Logistic regression uses a small stabilizing penalty; Lasso/Ridge/Elastic Net use 3-fold training CV.

Random-forest feature importance: top 6

Rank	Predictor	Raw importance	Share%
1	Business angel	127.31	18.87
2	Entrepreneurial intention	77.77	11.53
3	Age	54.22	8.04
4	Occupational status	44.45	6.59
5	Household size	36.99	5.48
6	Household income	31.63	4.69

Share is the percentage of total forest impurity importance across all 88 encoded predictors, not explained variance. Raw values are not comparable across samples. Missing indicators are derived preprocessing features.

Leading Legatum predictors within the same forest

Rank	Legatum predictor	Raw importance	Share%
1	Investment environment	4.16	0.62
2	Governance	3.90	0.58

Legatum subset ranked by raw importance; shares use total forest importance, not the Legatum subtotal.

Leading NES predictors within the same forest

Rank	NES predictor	Raw importance	Share%
1	Cultural and social norms	4.96	0.74
2	Higher/vocational entrepreneurship education	4.11	0.61
3	Government programs	3.59	0.53

NES shares use total forest importance.

Source: Completed Objective 5 NES extension batch, 29 September 2026; 02_obj5_india_iv_cv_nes_EXIT_ENT_harmonized. Scores and importance are taken directly from saved CSV outputs.